Diya Patel
24070126211

Assignment-8

Java/Experiment-8 at main · diyaa-13/Java

Experiment-8 > CollectionsDesignPatternApp.java > CollectionsDesignPatternApp

```java
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Comparator;
import java.util.Scanner;

class Product {
    int productId;
    String name;
    double price;
    int quantity;

    Product(int id, String name, double price, int qty) {
        this.productId = id;
        this.name = name;
        this.price = price;
        this.quantity = qty;
    }

    @Override
    public String toString() {
        return String.format("| %-4d | %-20s | %8.2f | %-5d |",
                productId, name, price, quantity);
    }
}

// Singleton Product Manager
class ProductManager {
    private static ProductManager instance = null;
    private ArrayList<Product> productList;
    private HashMap<Integer, Product> productMap;

    private ProductManager() {
        productList = new ArrayList<>();
        productMap  = new HashMap<>();
    }

```

Experiment-8 > CollectionsDesignPatternApp.java > CollectionsDesignPatternApp

```java
class ProductManager {

    public static ProductManager getInstance(){
        if (instance == null) instance = new ProductManager();
        return instance;
    }

    void addProduct(Product p) {
        if (productMap.containsKey(p.productId)) {
            System.out.println("Product ID already exists!");
            return;
        }
        productList.add(p);
        productMap.put(p.productId, p);
        System.out.println("Product added: " + p.name);
    }

    void removeProduct(int id) {
        Product p = productMap.remove(id);
        if (p != null) {
            productList.remove(p);
            System.out.println("Removed: " + p.name);
        } else {
            System.out.println("Product ID not found.");
        }
    }

    void displayProducts() {
        if (productList.isEmpty()) {
            System.out.println("No products.");
            return;
        }

        System.out.println("\n+------+----------------------+----------+-------+");
        System.out.println("| ID   | Name                 |  Price   | Qty   |");
        System.out.println("+------+----------------------+----------+-------+");
```

Experiment-8 > CollectionsDesignPatternApp.java > CollectionsDesignPatternApp

```java
class ProductManager {
    void displayProducts() {
        for (Product p : productList) {
            System.out.println(p);
        }

        System.out.println("+------+---------------------+----------+-------+");
    }

    void searchProduct(String keyword) {
        boolean found = false;

        for (Product p : productList) {
            if (p.name.toLowerCase().contains(keyword.toLowerCase())) {
                System.out.println("Found: " + p);
                found = true;
            }
        }

        if (!found) {
            System.out.println("No product matched: " + keyword);
        }
    }

    void sortByPrice() {
        productList.sort(Comparator.comparingDouble(p -> p.price));
        System.out.println("Products sorted by price.");
    }
}
```

Experiment-8 > CollectionsDesignPatternApp.java > CollectionsDesignPatternApp

```java
public class CollectionsDesignPatternApp {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ProductManager pm1 = ProductManager.getInstance();
        ProductManager pm2 = ProductManager.getInstance();

        System.out.println("Same instance? " + (pm1 == pm2));

        // Sample data
        pm1.addProduct(new Product(1, "Laptop", 65000, 10));
        pm1.addProduct(new Product(2, "Mouse", 800, 50));
        pm1.addProduct(new Product(3, "Keyboard", 1500, 30));
        pm1.addProduct(new Product(4, "Monitor", 18000, 8));

        while (true) {
            System.out.println("\n1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit");
            System.out.print("Choice: ");

            int ch;
            try {
                ch = Integer.parseInt(sc.nextLine().trim());
            } catch (Exception e) {
                System.out.println("Invalid input!");
                continue;
            }

            switch (ch) {
                case 1:
                    try {
                        System.out.print("ID: ");
                        int id = Integer.parseInt(sc.nextLine().trim());

                        System.out.print("Name: ");
                        String nm = sc.nextLine().trim();
```

Experiment-8 > CollectionsDesignPatternApp.java > CollectionsDesignPatternApp > main(String[])

```java
public class CollectionsDesignPatternApp {
    public static void main(String[] args) {
                        System.out.print("Price: ");
                        double pr = Double.parseDouble(sc.nextLine().trim());

                        System.out.print("Qty: ");
                        int qty = Integer.parseInt(sc.nextLine().trim());

                        pm1.addProduct(new Product(id, nm, pr, qty));
                    } catch (Exception e) {
                        System.out.println("Invalid input.");
                    }
                    break;

                case 2:
                    System.out.print("Product ID to remove: ");
                    try {
                        pm1.removeProduct(Integer.parseInt(sc.nextLine().trim()));
                    } catch (Exception e) {
                        System.out.println("Invalid!");
                    }
                    break;

                case 3:
                    pm1.displayProducts();
                    break;

                case 4:
                    System.out.print("Search keyword: ");
                    pm1.searchProduct(sc.nextLine().trim());
                    break;

                case 5:
                    pm1.sortByPrice();
                    pm1.displayProducts();
                    break;

                case 0:
                    System.out.println("Goodbye!");
                    sc.close();
                    return;

                default:
                    System.out.println("Invalid option.");
            }
        }
    }
}
```

Output:

```
PS C:\Users\vasum\Desktop\Diya\Java\Java\Experiment-8> java CollectionsDesignPatternApp
Same instance? true
Product added: Laptop
Product added: Mouse
Product added: Keyboard
Product added: Monitor

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 1
ID: 5
Name: CPU
Price: 20000
Qty: 2
Product added: CPU

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 2
Product ID to remove: 1
Removed: Laptop

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 3

+------+----------------------+----------+-------+
| ID   | Name                 |  Price   | Qty   |
+------+----------------------+----------+-------+
| 2    | Mouse                |   800.00 | 50    |
| 3    | Keyboard             |  1500.00 | 30    |
| 4    | Monitor              | 18000.00 | 8     |
| 5    | CPU                  | 20000.00 | 2     |
+------+----------------------+----------+-------+

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 4
Search keyword: Moni
Found: | 4    | Monitor              | 18000.00 | 8     |
```

```
PS C:\Users\vasum\Desktop\Diya\Java\Java\Experiment-8> java CollectionsDesignPatternApp

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 5
Products sorted by price.

+------+----------------------+----------+-------+
| ID   | Name                 |  Price   | Qty   |
+------+----------------------+----------+-------+
| 2    | Mouse                |   800.00 | 50    |
| 3    | Keyboard             |  1500.00 | 30    |
| 4    | Monitor              | 18000.00 | 8     |
| 5    | CPU                  | 20000.00 | 2     |
+------+----------------------+----------+-------+

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 6
Invalid option.

1.Add  2.Remove  3.Display  4.Search  5.SortByPrice  0.Exit
Choice: 0
Goodbye!
PS C:\Users\vasum\Desktop\Diya\Java\Java\Experiment-8>
```